

Mathanraj Rajendran

Research Assistant
Smart Construction & Systems Laboratory
Chungbuk National University, South Korea
Email: mathan@chungbuk.ac.kr
Alternative Email: mathan@skku.edu
Profile: <https://razmathan.github.io/>

EDUCATION

Master of Science in Global Smart City 2023-2025
Sungkyunkwan University, South Korea

- Thesis Title: Dimensional quality assessment of structural steel members using terrestrial laser scan data
- Advisors: Professor Sung-Han Sim, and Professor Min-Koo Kim
- GPA: 4.31/4.5 (Equivalent to ~3.83/4.0)

Bachelor of Engineering in Civil Engineering 2014-2018
Anna University, India

- Thesis Title: Strength and durability assessment of concrete with partial replacement of coarse aggregate by recycled plastic aggregate
- GPA: 7.3/10.0 (Graduated in First Class)

RESEARCH INTERESTS

- Scan-to-BIM and Digital Twin for construction and infrastructure management
- AI-driven task automation for construction workflows
- Automated inspection and condition assessment using multi-modal data analysis
- Construction robotics for autonomous monitoring systems

PUBLICATIONS

Journals:

1. Duorinaah, F. X., **Rajendran, M.**, Kim, T. W., Kim, J. I., Lee, S., Lee, S., & Kim, M. K. (2025). Human and Multi-Robot Collaboration in Indoor Environments: A Review of Methods and Application Potential for Indoor Construction Sites. Buildings, 15(15), 2794. 2025
<https://doi.org/10.3390/buildings15152794>
2. **Rajendran, M.**, Sim, S. H., Kim, M. K., Choi, Y. K., & Yi, C. Y. (2025). Automated dimensional quality assessment of steel members using TLS and enhanced edge line estimation. Measurement, 118990. 2025
<https://doi.org/10.1016/j.measurement.2025.118990>

Conference:

1. Ayembilla, F.B., **Rajendran, M.**, Kim, M. K., Semantic Segmentation of MEP Point Cloud Scenes Using Point Cloud Completion Approach, ICCEPM 2025, Jeju. <https://doi.org/10.6106/ICCEPM.2025.0306> 2025
2. **Rajendran, M.**, Sim, S. H., Kim, M. K., & Choi, Y. K. (2024). Dimensional Quality Assessment of Steel H-Beams Using Terrestrial Laser Scan Data. In International conference on construction engineering and project management (pp. 264-270). Korea Institute of Construction Engineering and Management. <https://doi.org/10.6106/ICCEPM.2024.0264> 2024

Under review:

1. **Rajendran, M.**, Duorinaah, F. X., & Kim, M. K. Advancing Indoor Construction with Autonomous Robots: Exploring Mobile Robot Applications, Operations, and Challenges in Building Environments. Journal of Information Technology in Construction. *(Under review)* 2025
2. **Rajendran, M.**, Ayembilla, F.B., Sim, S. H., & Kim, M. K., Automated Shape-Aware Dimensional Classification and Semantic-Instance Segmentation of Steel Structures from Occluded Point Cloud Data for Scan-to-BIM. Automation in Construction *(Under review)* 2026
3. Ayembilla, F.B., **Rajendran, M.**, & Kim, M. K., Improving Cross-Facility Generalization of MEP Semantic Segmentation Using Independent-Source Synthetic Point Cloud Data. Automation in Construction *(Under review)* 2026

RESEARCH EXPERIENCE

Research Assistant

2026-Present

Smart Construction & Systems Lab, Chungbuk National University ([Site](#))

Advisor: Prof. Min-Koo Kim

- Developing semantic segmentation framework for indoor point clouds using foundation models and Vision Language Models (VLMs)
- Developed semantic and instance segmentation pipeline for steel structures using synthetic point cloud data and deep learning-based occlusion reconstruction techniques.
- Developed synthetic data generation methods and AI-assisted real-time object detection for robotic bin picking applications.

Research Assistant

2025-2025

Built Environment Research Center, Sungkyunkwan University ([Site](#))

Advisor: Prof. Min-Koo Kim

- Investigated technical challenges affecting mobile robot operation in confined indoor construction environments.
- Studied human-robot collaboration strategies and evaluated their applicability and limitations in indoor construction processes.

Graduate Research Assistant

2023-2025

Smart Construction Laboratory, Sungkyunkwan University

Smart Construction & Systems Laboratory, Chungbuk National University

Advisors: Prof. Sung-Han Sim, and Prof. Min-Koo Kim

- Developed an automated method for accurate cross-sectional level detailed dimensional assessment of structural steel using terrestrial laser scanning point cloud data.
- Proposed a precise edge line estimation method to estimate actual edges of target object while using low-resolution TLS point clouds.

PROJECTS**Semantic Segmentation of Indoor Point Clouds**

2026-Present

- Developing zero-shot semantic segmentation methods for indoor point clouds using Vision-Language Models and image and point cloud fusion.

Bin Picking Using Robotic Arm Manipulation

2025-Present

- Developed real-time bin-picking systems using robotic arms, computer vision, and multi-modal sensing.

Dimensional Quality Assessment of Precast Concrete Panels

2023-2025

- Developed point cloud and image fusion methods for automated dimensional quality assessment of precast concrete components.

INDUSTRY EXPERIENCE**3D Modeler**

2020-2022

Rendersoft Systems India Pvt. Ltd., Chennai, India

- Developed detailed BIM models of buildings from architectural drawings using AutoCAD, 3ds Max, and Blender.

CAD Trainer

2018-2020

CADD Centre Training Services, Chennai, India

- Delivered training in 2D/3D CAD modeling and construction scheduling software, and supervised student design projects.

HONORS AND AWARDS

- Brain Korea Research Scholarship (BK21) by the Government of South Korea 2023-2025
- STEM Scholarship by Sungkyunkwan University 2023-2025

SKILLS

- **Programming:** MATLAB, Python, Rhinoceros scripting, and Dynamo visual programming for algorithm development
- **Computer Vision & AI:** Image and point cloud processing; deep learning applications; synthetic image and point cloud training data generation through scan simulations
- **3D Modeling & Visualization:** Revit, 3Ds Max, and Blender for modeling and visualization tasks
- **Simulation & Robotics:** Webots software for robot-environment simulation
- **Reality Capture:** Handheld and terrestrial laser scanning
- **Aerial Imaging:** Drone operation (DJI Mavic series) for data collection and photogrammetry-based modeling

LANGUAGE PROFICIENCY

IELTS Academic Score: Band 7/9

2026

ONLINE EDUCATION

- Python for Everybody, Coursera ([Cred](#))
- 3D Modeling in Rhinoceros with Python/Rhinoscript, Coursera ([Cred](#))
- Master Reality Capture with 3D Scanning, LinkedIn Learning ([Cred](#))
- Photogrammetry for product design and AEC, LinkedIn Learning ([Cred](#))
- Autodesk ReCap Pro: Processing Drone Images, LinkedIn Learning ([Cred](#))
- AR Development Techniques, LinkedIn Learning ([Cred](#))
- FARO Laser Scanning, LinkedIn Learning ([Cred](#))
- Matterport 3D Scanning and Visualization, LinkedIn Learning ([Cred](#))

REFERENCES

- **Dr. Sung-Han Sim** (Professor)
School of Civil, Architectural Engineering and Landscape Architecture
Sungkyunkwan University
Relationship: Thesis Advisor
Email: ssim@skku.edu
- **Dr. Min-Koo Kim** (Associate Professor)
Department of Architectural Engineering
Chungbuk National University
Relationship: Thesis Co-advisor
Email: joekim@chungbuk.ac.kr